

Load Interconnection Request System Impact Study Report

For Queue Project Q11-0061

Candelaria 138 kV
December 19, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0061. The Interconnection Customer proposes a 220.0 MW large load (data center) facility interconnecting at the Candelaria 138 kV substation in the Virginia area. The request was received on July 25, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 54 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$34,978,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Monel-Candelaria 138 kV	Bear-Clover 345 kV	136.5	241	44.0
Desert Outpost-Monel 138 kV	Brush Tavern-Rustburg 138 kV	122.6	232	30.7
Bluff City-Walworth 138 kV	Fluvanna P.S-Louisa Generation Facility 345 kV	116.1	309	11.5
South Texa-Desert Outpost 138 kV	Brush Tavern-Rusburg Substation 138 kV	112.6	319	22.4
Monel-Rustburg Substation 138 kV	Bear-Clover 345 kV	105.5	292	11.1
Brush Tavern-Rustburg 138 kV	Clover-Bear 345 kV	103.4	331	11.1
Reusens 138/1 kV TX	Bear-Clover 345 kV	101.3	519	6.0
Yankee Doodle-Red Hill 138 kV	Monel-Candelaria 138 kV	100.7	233	35.9
Otter-South Colton 138 kV #1	S Colton-Otter 138 kV Circuit 2	100.2	471	8.2
S Colton-Otter 138 kV	S Colton-Otter Substation 138 kV #1	100.2	471	8.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Monel-Candelaria 138 kV	Rebuild line with higher-capacity conductor	\$43,419,000	53% (\$23,028,000)
Desert Outpost-Monel 138 kV	Rebuild line with higher-capacity conductor	\$5,667,000	55% (\$3,133,000)
Bluff City-Walworth 138 kV	Rebuild line with higher-capacity conductor	\$4,596,000	100% (\$4,596,000)
South Texa-Desert Outpost 138 kV	Rebuild line with higher-capacity conductor	\$2,386,000	28% (\$663,000)
Monel-Rustburg Substation 138 kV	Reconductor line	\$6,383,000	15% (\$975,000)
Brush Tavern-Rustburg 138 kV	Reconductor line	\$4,718,000	15% (\$721,000)
Reusens 138/1kV Autotransformer	Replace transformer with higher-rated unit	\$6,766,000	14% (\$959,000)
Yankee Doodle-Red Hill 138 kV	Reconductor line	\$945,000	58% (\$551,000)
Otter-South Colton 138 kV #1	Reconductor line	\$2,253,000	8% (\$191,000)
Otter-S Colton Substation 138 kV	Reconductor line	\$1,894,000	8% (\$161,000)
	Total allocated to project		\$34,978,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.